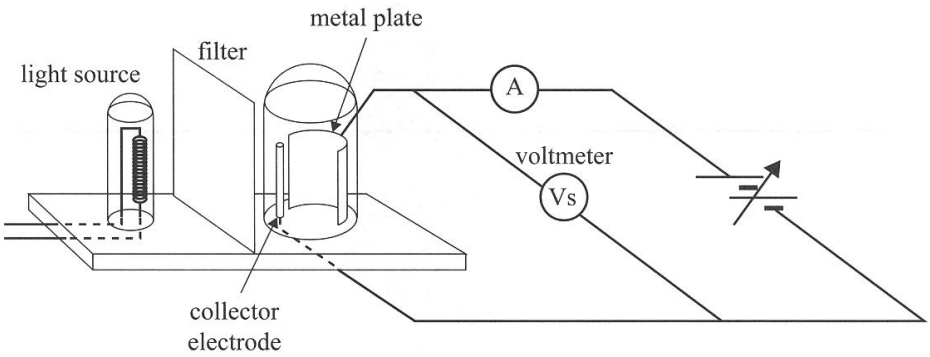
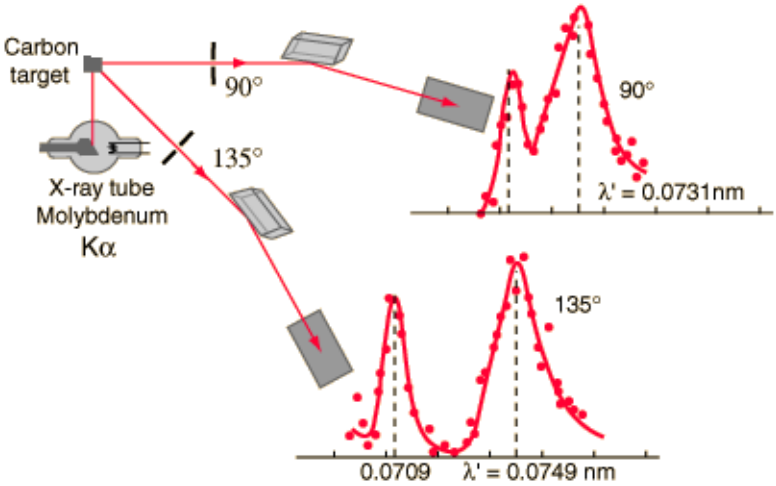
Discussion Questions

1 (a) A (clean) zinc plate is charged electrostatically. When ultraviolet light strikes the surface, the zinc plate discharges, even though it is on an electrically insulated surface. Explain how this occurs.
(b) Is it possible to charge an uncharged zinc plate by shining UV light on it?

2 In a typical photoelectric cell, as shown below, it appears that the circuit is incomplete, but a current can be observed to flow between the cathode and the anode. Explain how this can occur. (Include in your explanation the role of the battery in the circuit.)



3 X-rays of wavelength 70pm are scattered off a carbon target as shown in the figure. The resulting x-rays detected are plotted by their wavelength and relative signal intensity.



Explain why there are *two* peaks in each of the spectra.

4 Which is true? The minimum frequency required to observe the photo-electric effect is:
(a) The frequency that shakes the electron hard enough to remove it from the atom.
(b) The frequency that provides photons of just sufficient energy to overcome the binding energy of the electron to the surface.
(c) The frequency that provides sufficient amplitude to shake the electron loose.

5 Which is true? The Compton Effect is evidence that photons carry:
(a) energy and charge
(b) energy and momentum
(c) energy and mass

6 Compton scattering of photons is described by $\Delta\lambda = (h/m_e c)(1 - \cos\varphi)$. This shift is easy to observe with X – rays, but not with visible light. Why might this be the case?

7 If an electron and proton have the same kinetic energy, which then has the greater de Broglie wavelength?

8 What *non-classical* basic assumptions are made in the Bohr model of the hydrogen atom?

9 What does Bohr's correspondence principle say about quantum mechanics versus classical mechanics?

Problem-solving Questions

10 Monochromatic-light is incident on a polished zinc plate that forms part of a photoelectric measurement apparatus. Electrons are observed to be emitted from the surface of the plate, with maximum kinetic energy of 1.13 eV. The work function for zinc is 4.24 eV.

a) Calculate the energy (in eV) of photons in the incident light
b) Explain what is meant by the term *work function*.
c) What is the minimum kinetic energy of electrons which are emitted?
d) If the wavelength of the incident light was doubled, would you still expect to see electrons emitted from the plate? Give reasons for your answer.
e) Calculate the wavelength of the light.
f) If the intensity of the light was increased by 50%, by what fraction would the maximum velocity of the electrons increase?

11 X-rays of energy 40 keV are incident on a block of carbon and undergo Compton scattering.

(a) Calculate the wavelength of the incident X-rays
(b) What is the maximum possible Compton shift, and for what angle does this occur?
(c) Compton scattering in carbon occurs from both free electrons and integrated charge of the atom. How would you distinguish X-rays scattered by the atom from those scattered by free electrons?

12 A small laser is emitting light of wavelength 550 nm. It has a power of 1 mW. Calculate how many photons leave the laser every second.

13The following table gives the charge seen by Millikan at different times on single oil drops in his experiment. From the data, calculate the elementary charge *e*.

6.563×10^{-19} C	13.13×10^{-19} C	19.71×10^{-19} C
8.204×10^{-19} C	16.48×10^{-19} C	22.89×10^{-19} C
11.50×10^{-19} C	18.08×10^{-19} C	26.13×10^{-19} C

Halliday, David. *Fundamentals of Physics Extended, 10th Edition*. John Wiley & Sons,

14 Consider a planet orbiting a star. Most astronomical bodies can be considered 'black bodies'.

DATA
- radius of planet = 6.4×10^6 m
- solar radiation striking the planet = 1500 J m^{-2}. s^{-1}
- Constant in Stefan's Law (σ) = 5.7×10^{-8} W m^{-2} K^{-4}

(i) Show that the total energy reaching the planet in one second is 1.93×10^{17} J.

(ii) Now assume that the planet is in thermal equilibrium, and it radiates this amount of energy into space every second. You can assume that the planet is a 'black body', and that it radiates uniformly all over its surface.
Calculate the temperature of the planet in degrees Celsius.

(iii) Use your answer to part (ii) to estimate the peak wavelength (λ_{MAX}) radiated by the planet.

15 According to the Bohr model of the atom, the energies of an electron in a 'hydrogen-like' atom[1] are given by:

$$E_n = -\left(\frac{Z^2 m e^4}{8h^2\varepsilon_0^2 n^2}\right)$$

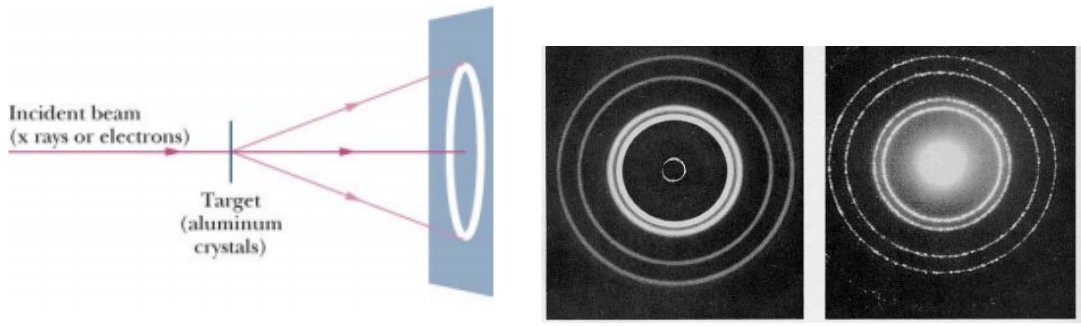
where n is the principal quantum number, and Z is the atomic number of the atom.

a) Calculate the value of $\left(\frac{m_e e^4}{8h^2\varepsilon_0^2}\right)$ in units of eV [Hint: simplify the powers first].

b) Calculate the longest wavelength of a photon of visible light emitted in the Balmer (n = 2) series of hydrogen.

c) An atom of helium has Z = 2. One of its electrons is removed. It is now He^+. These ions can also be raised to excited states, and in de-exciting to lower states, also emit radiation. Its spectral series can be called by the same three names as for the H spectra.
Calculate the *lowest* energy of a photon emitted in the Balmer series of a He^+ ion? What part of the electromagnetic spectrum will this photon be located in (visible, UV, infrared, etc)?

16 A beam of x-rays of wavelength 30 pm is directed onto an aluminium target as shown in figure (a), and the distribution of the scattered X-rays is shown in diagram (b) below.



(a) *(b)* *(c)*

If a beam of electrons of a suitable energy was used instead of X-rays, the distribution of the scattered electrons will resemble diagram (c). As can be seen, the two diffraction images have almost identical spacings.

a) What conclusion can be drawn from the similarity of the scattered X-ray and electron images?
b) What is the momentum of the electrons used in the electron measurement?
c) What is the energy of the electron beam?

17 X-rays of unknown wavelength are scattered from a crystal. The spacing between each atom in the crystal is 0.30 nm. A detector is used to measure the intensity of scattered X-rays at various angles, and a maximum is observed at an angle of θ = 30°.

a) Show that the path-difference between the scattered x-rays from adjacent layers is given by $2d\sin\vartheta$.

b) Calculate at least *two* possible values of the wavelengths of the scattered X-rays.\

[1] Such as He^+, Li^{++}……U^{91+}…..

18 Inside an x-ray machine, electrons are accelerated into a Tungsten target using a 40 kV potential difference, to produce x-rays. Calculate the shortest possible wavelength that can be emitted from its target.

19 The quantity $\alpha = \dfrac{e^2}{2\epsilon_0 hc}$ is sometimes described as the *fine structure constant*. Calculate its value in S.I units, and state its unit.

Past exam questions

20 X-rays of energy 56 keV (wavelength 22 pm) are Compton scattered from a carbon target, and the scattered x-rays are detected at an angle of 85° to the incident beam.

(a) Calculate the **wavelength** for x-rays that are Compton scattered by the outer-shell (essentially "free") electrons of the carbon atoms.

The carbon target is replaced with a much heavier target made of lead.

(b) What **now** is the wavelength for x-rays that are Compton scattered by the outer shell electrons of the lead atoms?

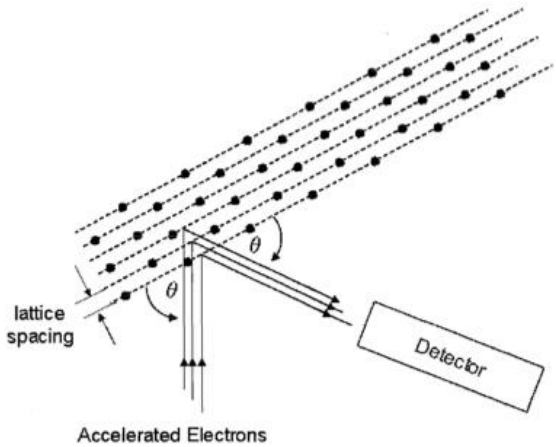
21 Australian freeway lighting is generally provided by sodium discharge lamps. These emit a yellow light of wavelength 589 nm. Each lamp typically consumes 750 W of electricity.

a) Calculate the energy of a single photon from such a lamp.
b) How many photons are emitted each second, assuming the lamp is 100% efficient.

22 A beam of low energy electrons is directed onto a crystal target, to study the angular dependence of the scattered electrons. This is similar to the setup used by Davisson and Germer where they found at particular angles there was a peak in the intensity of the scattered electron beam.

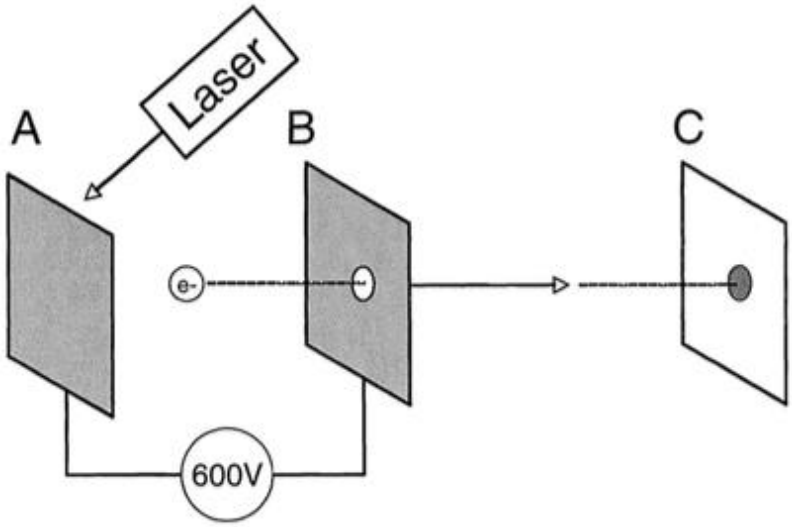
At an accelerating potential difference of 55 V, a definite peak in the scattered electrons is found at an angle of $\theta = 65°$.

a) Briefly explain the scattering process that leads to the formation of peaks in the intensity of scattered electrons. Include in your answer a reference to the property of electrons that is relevant.
b) Show that the momentum of the incident electrons is approximately $4.0\times10^{-24}\ \mathrm{kg.ms^{-1}}$
c) With the aid of Bragg's law ($2d\sin\theta = n\lambda$) and assuming n = 1, determine the lattice spacing in the crystal.



23

a) The diagram shows a simple electron gun setup for an electron microscope. It consists of a visible laser that is used to release electrons from a polished metal surface (A). These electrons are then accelerated from rest by a potential difference of 600 V between plates A and B. The geometry is such that the energetic electrons pass horizontally through a 1 mm hole in B and reach a phosphor screen C, where a bright spot of about 1 mm in diameter can be observed. The entire apparatus is operated under vacuum conditions.



i) Identify and explain the physical process by which the laser releases electrons from the metal surface.

ii) Estimate the velocity of the electrons as they pass through the hole at B.

iii) Calculate the de Broglie wavelength of the electrons.

In an attempt to reduce the spot size at C, the hole in the plate B is gradually made smaller and smaller. It is now almost as small as the de Broglie wavelength of the electrons.

b) Comment on whether this procedure will achieve the desired goal. Justify your answer.

24

a) Explain what is meant by the Heisenberg Uncertainty Principle $\left(\Delta x \Delta p \geq \frac{\hbar}{2}\right)$.

In the Bohr model, an electron in the ground state of a hydrogen atom is considered to have a kinetic energy of 13.6 eV.

b) Show that the momentum of this electron is approximately $2\times10^{-24}\ \mathrm{kg.ms^{-1}}$.

c) Use the Uncertainty Principle to estimate the diameter of the hydrogen atom.

25

a) In classical physics an atom would not be stable because it would emit electromagnetic waves of frequency *f*, because the electron's orbit, when seen edge on, appears to be an electric dipole. At what electron radius, in nm, would the electron emit light with a wavelength of 600 nm?

b) We wish to measure the wavelength and position of a photon simultaneously. The wavelength of the photon is 600 nm and is known with an accuracy $\frac{\Delta p}{p} = 10^{-6}$. What is the minimum uncertainty in the position of the photon?

c) An electron is confined between two impenetrable walls 0.2 nm apart. Determine the energy levels for the states n =1, 2 and 3.

Physics 2: Physical Science & Technology (PHYC10004) SUM 2025

Short solutions

Discussion Questions:

1a This only occurs if the zinc plate is negatively charged. The UV light photons have enough energy to release electrons from the surface (if it is clean). They are then repelled away from the plate.
1b Electrons are released, but the Zn plate is now positively charged, and they are attracted back to the plate
2 electrons are released from the metal plate and move across to the electrode; this current completes the circuit; the battery provides a stopping voltage to prevent current below certain kinetic energies.
3 One is due to elastic scattering of the x-ray off the target nucleus (0.0709 nm), the other is due to inelastic scattering off outer/loosely bound electrons (0.731/0.0749 nm).
4 (b) **5** (b) **6** Compton shift is still there but is *very* small (pm) compared to the wavelength of light (hundreds of nm) and hence difficult to detect
7 electron de Broglie wavelength is greater as $\lambda = \frac{h}{mv}$
8 angular momentum is quantised (leads to quantisation of energy and orbit radius); orbiting (accelerating) electrons do not radiate energy; electrons can jump from one stationary state to another, emitting a photon
9 At large quantum numbers, predictions of quantum mechanics = classical mechanics.

Problem Solving Questions:

10a 5.37 eV **c** 0 eV (only just escaping) **d** Probably not; energy of incident photons would halve; would be below the work function. **e** 231 nm **f** would not change
11a 0.031 nm **b** occurs at 180 degrees ; equal to $2h/mc$ = 4.85 pm scattering off electrons)
c Those scattered from whole atoms have Compton shifts almost equal to zero.
12 2.8×10^{15}
13 First, find an *approximate* value of *e* by taking differences between all the given data. Now assign integers *n* using this approximate value of *e* (4.1*e* is clearly 4*e,* 5.13*e* is 5*e*, and so on). Now calculate an *e* value for each of these, and average. You should get: 1.641×10^{-19} C.
14i Take half the planet's surface area (facing its star) and incident radiation value, gives power (Watts=J/s), multiply by 1 second.
ii Assume that the planet is in thermal equilibrium and use Stefan's Law; T = 12°C
iii Use Wien's Law; 1.02×10^{-5} m
15a 13.6 eV **b** 656 nm (this is 1.89 eV) **c** 7.6 eV (this is UV)
16a same wavelengths **b** 2.2×10^{-23} N s **c** 1.7 keV;
17b 3×10^{-10} m (*n = 1)* and 1.5×10^{-10} m (*n* = 2) etc…
18 max KE transferred will be 40 keV, set equal to hc/λ; answer is 31 pm.
19 value is 7.3×10^{-3}; it has no units (to check, can reduce everything to SI)

Past Exam Questions:

20 24.2 pm (both parts)
21 3.37×10^{-19} J =2.11 eV and 2.22×10^{21} photons/second
22(c) $9.14 \times^{-11}$ m
23ai photoelectric effect **aii** 1.45×10^{7} m/s **aiii** 5×10^{-11} m
b there will now be significant diffraction, spreading the spot

24a see text Walker's *Principles of Physics*, 11th Ed., p1172 and following –"roughly... all physical quantities... are subject to unpredictable fluctuations, so that their values are not precisely defined" – Paul Davies, 1989. Important to note it's fundamental – not a failure of experimental technique. **b** use $E = p^2/2m$ **c** somewhere near 10^{-11} - 10^{-10} m
25a 2.95 Å **b** 0.047 m **c** 9.42 eV, 37.7 eV, 84.8 eV